Siddhi Kale

siddhi.kale2503@gmail.com 8530182828 https://www.linkedin.com/in/siddhikale25
https://github.com/siddhi-works

Summary

Computer Science student with foundational experience in software development and AI, seeking an internship. Developed full-stack capabilities with Node.js, TypeScript, and React.js, culminating in an AI-powered insurance claim assistant project. Proficient in API design, database management with SQLite, and collaborative development using Git/GitHub. Applied data structures and algorithms to resolve technical problems.

Experience

Content Editor & Management Coordinator

Apr 2025 - Nov 2025

Abhivyakti Editorial Team, ADYPU

Pune, Maharashtra, India

- Managed content calendar for 5 social media platforms, increasing engagement by 25% year-over-year.
- Coordinated with cross-functional teams to ensure timely delivery of 50+ content pieces per quarter.
- Edited and proofread all outgoing content, maintaining brand consistency and reducing errors by 15%.
- Implemented a new content workflow that improved production efficiency by 30%.

Content & Marketing Team Volunteer

Nov 2021 - Mar 2023

Bindu Non-Profit Organisation

Remote

- Managed a team of 5 content creators, increasing content output by 25% within the first quarter.
- Developed and executed a comprehensive content strategy that drove a 15% increase in organic traffic year-over-year.
- Oversaw the creation of all marketing collateral, including whitepapers, case studies, and social media campaigns, resulting in a 10% improvement in lead generation.
- Collaborated with sales teams to align content with marketing objectives and sales enablement needs.

Research Volunteer

Sep 2024 - Present

NSS, ADYPU

Pune, Maharashtra, India

- Supported the planning and execution of 8 youth programs, impacting over 100 children annually.
- Mentored and guided participants, fostering a positive and inclusive environment.
- Assisted in preparing program materials and resources, ensuring smooth session delivery.
- Managed participant registration and communication, maintaining accurate records for 100% of attendees.

Education

Bachelor of Technology in Artificial Intelligence and Data Science

Jul 2023 - Jul 2027

Ajeenkya Dy Patil University

Pune, India

- CGPA - 9.02/10

Projects

ClaimMate — AI-Powered Insurance Claim Companion

Developed an AI-powered insurance claim assistant using multi-agent AI orchestration to analyze claim context and policy details.

- Engineered a multi-agent system orchestrating specialized agents for policy, claim, and document analysis, demonstrating advanced API design and integration.
- Developed a robust backend reasoning pipeline to process user inputs, policy data, and agent outputs, maintaining claim context efficiently.
- Executed comprehensive document and claim analysis logic to identify inconsistencies and pending actions, improving claim processing.
- Developed and deployed a bilingual React + TypeScript web application with a Node.js/Express backend, integrating Gemini 2.5 Flash through secure server-side APIs.

Generative AI | LLM | Agentic AI | github | render | react | nodejs | typescript | RESTful APIs

Resolve IQ : AI-Powered Intelligent Ticket Routing & Resolution Agent

Developed an end-to-end IT ticket intelligence pipeline for automated classification, prioritization, routing, retrieval, and escalation.

- Applied data structures and algorithms, including TF-IDF and Logistic Regression, for automated ticket classification across IT support categories.
- Designed and implemented a rule-based intent detection system with machine learning fallback to enhance issue handling efficiency.
- Developed a knowledge retrieval module interacting with historical data to identify similar incidents and surface resolved solutions.
- Engineered confidence-based decision logic for priority estimation and recurring issue detection, supporting efficient ticket escalation.
- Developed Flask/Streamlit interfaces with SQLite persistence, supporting ticket lifecycle management, search/filtering, analytics, file attachments, and CSV/PDF reporting.

python | scikit-learn | logistic regression | pandas | numpy | flask | RESTful API | SQLite

ARIA : AI-Powered Research & Intelligence Agent

Developed an AI-powered research assistant that automates the end-to-end research workflow from retrieval to paper generation.

- Designed a multi-agent workflow using LangGraph and LangChain, orchestrating reasoning and tool execution for research automation.
- Integrated the arXiv API and a PDF analysis pipeline to retrieve, extract, and synthesize insights from academic literature.
- Automated the creation of structured LaTeX documents, generating publication-ready research papers exported as PDF reports.
- Developed a responsive Streamlit web application with real-time AI responses, persistent chat history, and downloadable research reports.

python | Langgraph | Langchain | Google Gemini 2.5 Flash API | LaTeX | arXiv API | Git & GitHub | PyPDF2 | streamlit

AI-Powered Dermoscopic Skin Lesion Classification and Detection

Developed a deep learning system using CNN (TensorFlow/Keras) for skin lesion classification and built a Streamlit web application.

- Developed a deep learning model for classifying dermoscopic skin lesions, achieving 95% accuracy on a 10,000-image dataset.
- Engineered image preprocessing pipelines to enhance feature extraction and optimize data quality for model training.
- Integrated the classification model into a web application, providing real-time diagnostic assistance.
- Validated model performance against expert diagnoses, demonstrating high concordance in classification.

Python TensorFlow Keras Scikit-learn Flask pycharm

AI-Based Chest X-Ray Disease Detection Using ResNet-50

Created an AI chest X-ray classification system with ResNet50 and transfer learning, integrated into a Flask web application.

- Developed and trained a ResNet-50 model for chest X-ray disease detection, achieving 92% accuracy on a diverse dataset.
- Utilized data augmentation techniques to enhance model robustness and generalization for medical imaging analysis.
- Preprocessed and curated a large-scale dataset of over 10,000 chest X-ray images, ensuring data quality.
- Fine-tuned pre-trained ResNet-50 layers to optimize performance for medical imaging tasks, applying machine learning principles.

Python TensorFlow Keras OpenCV Pandas NumPy Scikit-learn Matplotlib

Plagiarism Detection System Using NLP

Developed a plagiarism detection web application using NLP techniques, TF-IDF vectorization, and Cosine Similarity.

- Developed a Python-based plagiarism detection system, applying NLP techniques to analyze text similarity.
- Implemented natural language processing techniques, including TF-IDF and Cosine Similarity, for effective document comparison.
- Achieved 90% accuracy in identifying duplicated content across a dataset of 1,000 documents.
- Streamlined the text preprocessing pipeline for optimized performance and faster analysis.

Python NLTK Scikit-learn Pandas Flask streamlit

ML-Based Intelligent Bot Detection Model for Social Media

Built a Twitter bot detection system using R, Machine Learning, and NLP, performing feature engineering and training models.

- Developed a machine learning model using R to identify and mitigate bot activity on social media platforms.
- Engineered feature extraction pipelines using R packages to process large-scale user interaction data efficiently.
- Validated model performance, achieving 95% accuracy in detecting sophisticated bot networks.
- Reduced false positive rates by 60% through iterative refinement and hyperparameter tuning of the model.

R dplyr caret Shiny ggplot2 tidyr data.table

Technology

Programming Languages

Python R SQL

Web Technologies & Frameworks

React Node.js Flask Streamlit RESTful APIs

Databases

SQLite MySQL

Version Control

Git GitHub

Core Concepts & Methodologies

Problem Solving Analytical Thinking API Design Machine Learning Deep Learning Natural Language Processing Generative AI Large Language Models LLM Agentic AI

Libraries & Tools

TensorFlow Keras Scikit-learn NumPy Pandas NLTK OpenCV Matplotlib Langchain Langgraph PyPDF2 Google Gemini 2.5 Flash API arXiv API LaTeX PyCharm

Soft Skills

Teamwork Communication Leadership

Achievements

Published Research Paper – ICAIML-AE 2026

Published the research paper "Hybrid Feature-Based Comparative Analysis of Supervised and Unsupervised ML-Based Intelligent Bot Detection Model for Social Media Data" in the proceedings of the International Conference on Artificial Intelligence and Machine Learning for Agro-Ecosystems (ICAIML-AE 2026), organized by Vishwakarma University in collaboration with Binghamton University, USA, and published by Atlantis Press (Springer Nature).

Certificates

AWS re/Start Graduate

Amazon Web Services (AWS)

Tata - GenAI Powered Data Analytics Job Simulation

Forage

Data Science

CodeWithHarry

Artificial Intelligence and Machine Learning

Edunet Foundation

Languages

English

Hindi

Marathi